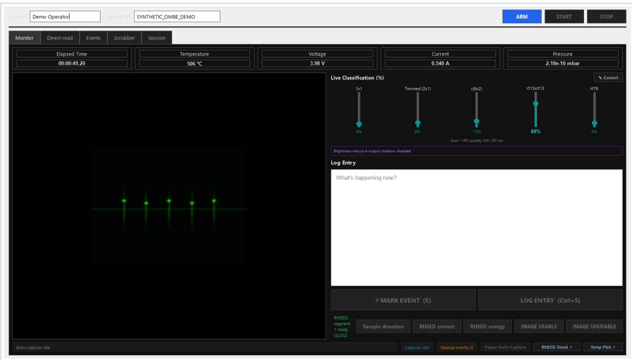
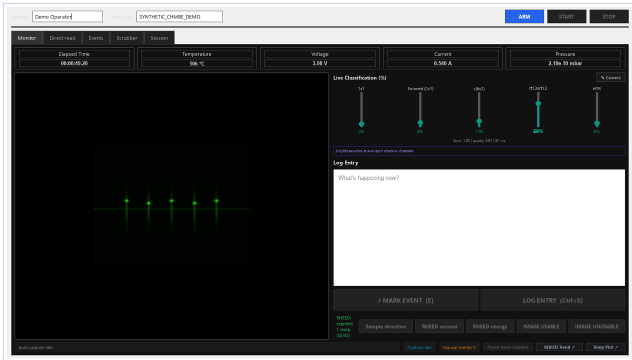
O-MBE + Ch-MBE GUI and RHEED Point-event Annotation

English Operator Manual



O-MBE Growth Monitor
Generated demo inputs; no hardware.



Ch-MBE Growth Monitor
Generated demo inputs; no hardware.

> The post-processing labeler reads archived data only. It does not connect to or control cameras, pyrometers, power supplies, or any other instrument.

Quick entry	Double-click
O-MBE live GUI	`Start O-MBE Growth Monitor.cmd`
Ch-MBE live GUI	`Start Ch-MBE Growth Monitor.cmd`
RHEED offline labeler	`Start RHEED Post-processing Labeler.cmd`
Install or refresh desktop shortcuts	`Install AI4MBE Desktop Shortcuts.cmd`
Open the English operator manual	**AI4MBE Operator Manual** shortcut
Remove installed program files	**Uninstall AI4MBE Growth Monitor** shortcut

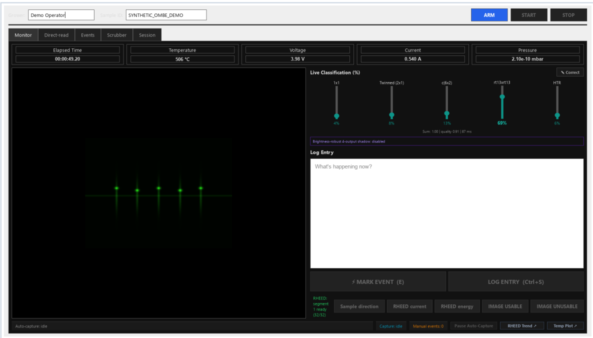
Manual v2.1 - English edition
Issued 2026-08-20 for the Windows O-MBE and Ch-MBE workstations.

The screenshots show the actual applications using generated demo inputs. No experimental dataset is included.

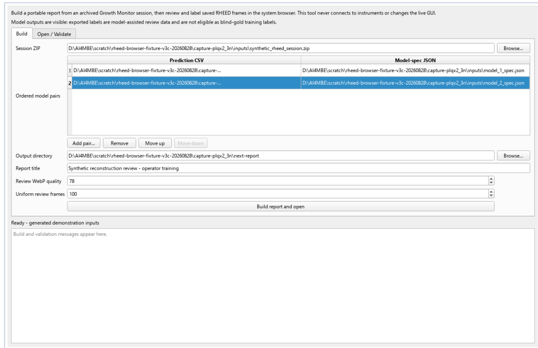
1 Understand the live and offline paths

In brief. The live GUI creates immutable session evidence. The offline labeler starts from the explicit 1x1 initial state, reviews manual, automatic-change, and posthoc point events, and derives complete state intervals with one Anchor each.

The live GUI performs acquisition and display. The offline labeler works only on completed archived sessions. Never interpret an offline result as a real-time control instruction.



1. Live acquisition
The chamber-selected Growth Monitor writes the archive.



2. Offline build and validation
The labeler combines the ZIP with predictions and model specifications.

The live GUI creates session records. The labeler reads those records later, produces a browser report, and exports validated JSON. It never sends commands back to the instruments.

What this manual covers

Chapter	Topic	Outcome
2	Chamber-specific live startup	Select the correct launcher and verify its exact title
3	Configuration modes and chamber defaults	Select the validated reader without cross-copying
4	Labeler desktop application	Select inputs, build, open, and validate
5-6	Inputs and safe report build	Produce a complete local report directory
7-8	Point events, interval Anchors, and display controls	Complete traceable reviews on actual saved frames
9	Export, import, and validation	Produce canonical JSON and validate fail-closed
10-12	Safety, troubleshooting, and version checks	Handoff and reproduce the work safely

Reconstruction event labels, pattern-clarity labels, image acquisition quality, model output, and FeSe film quality are different concepts. A reconstruction label records that `1x1`, Twinned `2x1`, `c(6x2)`, RT13, or HTR appeared or disappeared. A clarity label records that the visible pattern became Good or Bad. Image acquisition quality means only whether a captured frame can be analyzed; it never means Bad clarity, poor surface quality, or poor film quality. The current classifier concerns the bare STO surface before growth.

Operating principle

- The selected O-MBE or Ch-MBE GUI reads live instrument sources and writes chamber-identified session records under the operator's control.
- Post-processing combines an archived session with one or more already-generated prediction tables.
- Reviews are stored as frame-bound point events with immutable source evidence and append-only revisions.
- Every semantic label is editable and records either a reconstruction appearance/disappearance or a Good/Bad pattern-clarity change.
- Model outputs remain visible, so these annotations are assisted review, not blind-gold labels.

2 Install shortcuts and start the correct chamber GUI

In brief. Refresh the five desktop shortcuts, choose exactly one chamber launcher, verify its title and chamber preflight, and check live status before ARM. The launchers do not authorize operation or setpoint changes.

Shared first setup or after moving the checkout

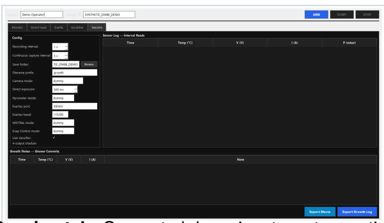
1. From the GUI repository root, double-click `Install AI4MBE Desktop Shortcuts.cmd`.
2. Confirm that the desktop contains exactly **O-MBE Growth Monitor**, **Ch-MBE Growth Monitor**, **RHEED Post-processing Labeler**, **AI4MBE Operator Manual**, and **Uninstall AI4MBE Growth Monitor**.
3. Confirm that the three application shortcuts show the bundled Yang Lab icon from `assets/ai4mbe_app_icon.ico`. The icon is visual identification only and does not prove chamber, repository, commit, or environment identity.
4. If Windows displays an unexpected security warning, stop and ask the maintainer to verify the file source, branch, and commit. Do not bypass it.

The three application shortcuts and root-level `Start *.cmd` wrappers route through `scripts/windows/launch_ai4mbe.ps1` with `-Application ombe`, `chmbe`, or `labeler`. It resolves a compatible 64-bit `ai4mbe-gui` Python, starts from the repository or installed-program root, sanitizes process-local Python and Qt variables, and writes diagnostics below `%LOCALAPPDATA%\AI4MBE\LauncherLogs`. It does not install packages, change persistent settings, or select acquisition modes; `AIQM_GUI_PYTHON` may identify an already validated interpreter. For either growth monitor it forces and verifies the requested chamber before Python can cache configuration, then owns a chamber-specific single-instance mutex. Runtime or chamber-preflight failure blocks launch. Optional-driver warnings are separate and remain stop conditions for production modes that require the missing driver.

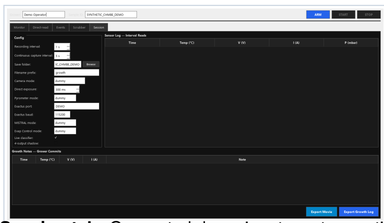
Select exactly one chamber branch

Physical chamber	Required launcher	Exact window title
O-MBE	`Start O-MBE Growth Monitor.cmd`	**Oxide MBE Growth Monitor**
Ch-MBE	`Start Ch-MBE Growth Monitor.cmd`	**Chalcogenide MBE Growth Monitor**

A different title or a launcher that does not match the physical chamber is a stop condition. Close normally and restart with the correct launcher; changing a GUI field is not a substitute.



O-MBE Session tab. Generated demo inputs; not operating defaults.



Ch-MBE Session tab. Generated demo inputs; not operating defaults.

Shared ARM and exit boundary

1. Confirm the title-selected chamber, advancing RHEED sequence, valid temperature state, and intended log directory. Check the four reader modes against the matching startup-default record in Chapter 3; verify all remaining operating settings against the applicable SOP.
2. ARM or start only when the authorized operator and applicable SOP permit it. Double-click startup authorizes no instrument operation or setpoint change.
3. Close normally and wait for the window to disappear. Do not force-kill Python while logs are being written.

3 Understand Configuration modes and chamber defaults

In brief. Both chambers start with `vimba`, `modbus`, `ads`, and `elog`: direct reads for RHEED, temperature, MISTRAL, and EvapControl. `dummy` modes generate test data and must not be mistaken for instruments.

The Configuration panel chooses one reader for each data source. A mode selects *how the GUI reads*; it does not change an instrument setpoint. The four selectors are independent, and their selected values are frozen when the session is armed.

Camera mode - RHEED image source

Option	Meaning
`dummy`	Fixed generated/demo STO 1x1 image; no camera connection.
`dummy_c6x2`	Fixed c(6x2) demonstration image; no camera connection.
`dummy_tw`	Fixed Twinned (2x1) demonstration image; no camera connection.
`dummy_rt13_tilted`	Rotated RT13 demonstration image for alignment testing; no camera connection.
`screengrab`	Windows Graphics Capture of a detached kSA *Live Video* window. It captures the window even when another window covers it, but it is still display pixels rather than raw camera data.
`screengrab_mss`	Legacy monitor-pixel capture. Window position, overlap, DPI, and foreground order can contaminate the frame; diagnostic fallback only.
`vimba`	Direct AVT camera acquisition through the Vimba SDK. This is the default live RHEED source. kSA camera ownership/access mode must permit the connection.

Vimba exposure and ARM fail-closed behavior

The **Direct exposure** control applies only to `vimba`. The current O-MBE profile requests 500 ms and the current Ch-MBE profile requests 300 ms during ARM. They are deliberately different software requests, not transferable chamber setpoints or operating authorization. The confirmed camera readback, not the displayed request alone, is the evidence of what was applied. The control is locked while armed or running.

- A nonzero request is a volatile camera write. It requires Full camera access and `ExposureAuto=Off`. Close kSA or Vimba X Viewer when either application prevents Full access.
- Zero displays **Keep current**. It performs no exposure write, although the driver still reads and records the current exposure when available.
- The UI and driver reject non-finite, out-of-range, or unsafe exposure and trigger combinations. The ceiling keeps at least 10 percent of the trigger period for acquisition overhead: 900 ms at the default 1 Hz rate.
- After writing, the driver reads the feature back. ARM succeeds only when the requested value can be proven applied. On a post-write failure, the driver attempts a verified restoration of the original exposure; an unverified restoration remains an explicit stop condition.
- The application never calls `UserSetSave`. The change is volatile, and a camera power cycle restores the stored camera user set.

Session metadata stores `camera_exposure_requested_ms` separately from `camera_exposure_readback_ms`. **Keep current** therefore has a null request and may still have a numeric readback. A request without a matching confirmed readback is not proof that the camera accepted it.

Camera ownership, freshness, and start gate

ARM is one cancellable startup transaction. A failed or abandoned setup releases the worker and camera ownership before the GUI returns to idle; never start a second GUI to work around an ARM failure. The direct Vimba path does not re-serve its cached last frame as a new acquisition. Capture sequence and arrival time must advance, and START remains disabled until the current ARM cycle delivers a qualifying live frame. If delivery stalls, DISARM and investigate camera ownership, triggering, sequence, age, and the recorded error rather than treating the last displayed image as current data.

Windows Graphics Capture can read a detached kSA *Live Video* window while another window covers it, but it fails closed if the source window is minimized, closed, or stops delivering new frames. `screengrab_mss` remains a monitor-pixel diagnostic fallback that can be contaminated by overlap, position, scaling, and display changes.

Pyrometer mode - substrate temperature source

Option	Meaning
`dummy`	Generated temperature values for offline GUI checks; not an instrument reading.
`exactus`	Direct serial read using the Exactus protocol and the displayed COM/baud fields.
`modbus`	Direct read from the pyrometer using the chamber profile's COM, baud, RTS, device ID, and Modbus backend. This is the default.
`screengrab`	Reads the existing TemperaSure application through Windows UI automation. It depends on the correct window being open and updating.

MISTRAL mode - power and cell data source

Option	Meaning
`dummy`	Generated voltage/current values; no PLC or MISTRAL connection.
`screengrab`	Captures the MistralGui window and OCRs setpoint and actual voltage/current. The window must be visible and the calibrated layout must match.
`jsonrpc`	Experimental HTTP direct connection. Its read-method map is not configured, so it can connect while returning no V/I values; do not use for production logging.
`ads`	Read-only direct Beckhoff TwinCAT ADS access using the chamber-specific endpoint, ports, schema, and cell count. This is the default.

EvapControl mode - pressure and evaporation data source

Option	Meaning
`dummy`	Generated pressure values; no EvapControl connection.
`elog`	Directly reads EvapControl's current `.elo` binary log. It needs no OCR or visible window and can expose pressure, substrate, cell, and plasma fields present in the validated schema.
`screengrab`	Captures the Evaporation Control window and OCRs chamber pressure. The window must be visible; fields not shown in that crop remain unavailable.

`vimba`, `modbus`, `ads`, and `elog` avoid screen OCR, but each worker still receives data independently. A sensor-log row is a latest-value software snapshot, not proof that exposure, temperature, PLC, and EvapControl were physically sampled at one instant.

O-MBE GUI startup defaults

Selector	Selection and source
Camera	`vimba` - direct AVT camera
Pyrometer	`modbus` - COM4, device 1, RTS off, `pymodbus`
MISTRAL	`ads` - read-only profile, 6 cells
EvapControl	`elog` - O-MBE `.elo` directory

Ch-MBE GUI startup defaults

Selector	Selection and source
Camera	`vimba` - direct AVT camera
Pyrometer	`modbus` - COM3, device 1, RTS off, `raw_serial`
MISTRAL	`ads` - read-only profile, 7 cells
EvapControl	`elog` - Ch-MBE `.elo` directory

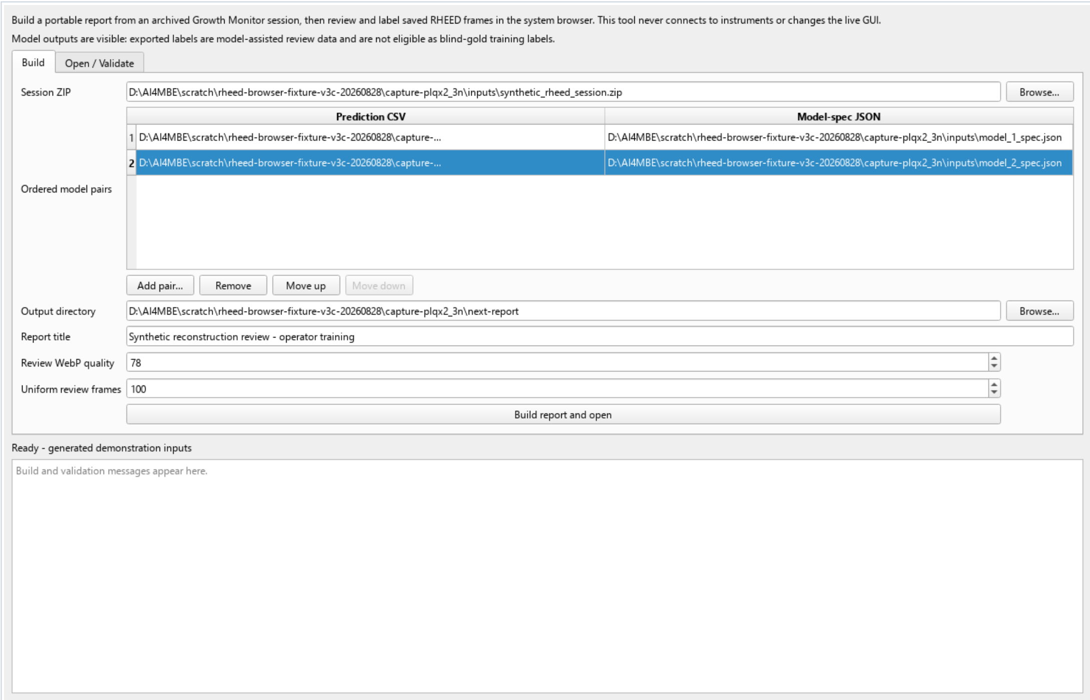
These are software startup selections, not permission to ARM. ROI/crop, classifier package and enable state, save-root approval and naming, sampling intervals, and any instrument setpoint remain governed by the matching chamber SOP and owner approval. The initial Windows save field uses separate `OMBE` and `ChMBE` folders and remains editable before ARM. Each production profile passes its own explicit directory to `ElogReader`; the generic `AIQM_EVAP_LOG_DIR` remains a standalone-reader fallback and cannot override the GUI profile. Session metadata records the effective pyrometer serial fields, configured Elog directory, and exact `.elo` source once connected. Variables absent from that chamber's schema remain blank rather than being invented.

4 Open the RHEED post-processing labeler

In brief. Use the desktop labeler for durable revisions and Complete. It remains offline from instruments and exposes only a random-token service on `127.0.0.1` for the selected local report.

Step 1. Double-click `Start RHEED Post-processing Labeler.cmd` or its desktop shortcut.

Step 2. Confirm that the application has the **Build** and **Open / Validate** tabs, plus **Build report and open**, **Open report**, **Validate annotations**, and **Open English PDF manual**. It must not start Growth Monitor or contact instruments.



Actual application screenshot. The current desktop labeler Build tab populated with a generated local fixture. The application is offline and no instrument process is running.

Area	Purpose
Session ZIP	Select one archived Growth Monitor session
Model inputs	Pair one prediction CSV with its model-spec JSON on each row
Output directory	Select a new or empty directory, preferably outside Git
Message log	Preserve complete build and validation messages
Open report button	Open a previously generated `interactive_report.html`
Validate annotations	Check exported JSON against the exact report provenance

This application can run on a computer without instrument software when the archived inputs are complete and the Python environment is valid. Report construction and validation run in an isolated child process so the desktop application remains responsive.

The desktop application serves the selected report only on the Windows loopback interface and gives it a random session token. The browser can then record audited revisions against the selected report. A report opened directly with `file:///` can navigate and edit a local Draft, but **Complete** remains unavailable and explains that the desktop labeler is required. Equalizer is a separate diagnostic and is not shown or requested in the event-labeling workflow.

5 Prepare inputs and verify provenance

In brief. Preserve the ZIP and exact saved frames. The explicit initial state, manual marks, automatic image-change candidates, and posthoc events become reviewable Drafts; adjustment, image-unusable, and instrument records remain read-only references.

Session archive

The ZIP archive must contain exactly one member ending in `session_metadata.json`, exactly one member ending in `heartbeat_log.csv`, and every sibling `frames/` image referenced by the heartbeat rows. Saved-frame elapsed time, heartbeat index, capture sequence, and timezone-aware capture UTC must each increase strictly; gaps are allowed. Do not extract and rename frames manually.

Read `chamber_id` from the session metadata and confirm that it names the intended O-MBE or Ch-MBE source. For Modbus, preserve COM, baud, RTS, device ID, and backend. For ADS, preserve endpoint, ports, and cell count. For Elog, preserve the configured directory and resolved source file. Compare provenance only with the corresponding startup-default record in Chapter 3; never edit an archive to make it resemble the other chamber.

```
growth_session.zip
  session_metadata.json
  heartbeat_log.csv
  frames/
    heartbeat_000001_....bmp
    heartbeat_000002_....bmp
```

Prediction tables and model specifications are positional pairs

File	Required content	Check
Prediction CSV	frame index, heartbeat index, elapsed time, capture UTC, capture sequence, frame name, frame SHA-256, and score columns	Rows match the saved-frame order exactly
Model-spec JSON	schema version 1, non-empty key and title, at least two unique classes, and equally sized unique probability columns	Class order matches the score columns; status and provenance are optional
Session ZIP	metadata, heartbeat log, and frames	Every referenced frame exists in the archive

Prediction `frame_index` may begin at 0 or 1, but it must then remain contiguous. Heartbeat indices, capture sequences, and elapsed times may contain gaps. The tool follows actual saved frames and recorded times; it never assumes exact 1 Hz sampling.

Any mismatch in row count, order, time, sequence, filename, or SHA-256 stops the build. Do not edit the inputs to bypass an error.

Event and context sources

Every run starts from the explicit state `1x1 present` with pattern clarity `unknown`. This is the starting state for replay, not a fabricated physical appearance event. Its audit item owns the first interval Anchor.

- `manual`: one-click **MARK EVENT** evidence created during acquisition; an empty comment remains Unfinished.
- `auto_capture`: a candidate point produced by the translation-insensitive RHEED image-change detector. It marks a visual change, not its physical cause, and must be Confirmed or Rejected.
- `posthoc`: a point added later at an actual saved frame.
- RHEED direction, beam-current, and beam-energy adjustments are read-only reference points.
- An image-unusable record means only that acquisition or capture made the image unsuitable for analysis. It is not a surface-quality or film-quality judgment.
- Temperature, voltage, current, pressure, and data age come from original logs and remain read-only context.

6 Build and open a report safely

In brief. Build only into a new or empty directory outside Git. Preserve the complete report directory and the original prediction/model-spec inputs separately.

Older `events_labels.csv` rows may seed automatic-event review. A legacy `live_labels.csv` row links only when capture sequence and frame SHA-256 identify exactly one event. Legacy temporal segments remain readable but are not converted automatically because a segment has no unique scientific point.

Recommended local layout

```
<local-data-root>\2026-08-11_anneal\
  source\growth_session.zip
  predictions\model_A.csv
  specs\model_A.json
  output\
  exports\
```

Step 1. Choose the preserved archive copy in Session ZIP.

Step 2. Add one Prediction CSV and Model-spec JSON row for every model to display. Pairing is positional, so preserve row order.

Step 3. Choose a new or empty directory outside the repository. The desktop labeler will not overwrite a non-empty directory.

Step 4. Enter a report title. Review quality changes only the lossy review images; it never changes archived pixels or model predictions. The default value of 78 is suitable for routine use.

Step 5. Click **Build report and open**. Wait for success and for `interactive_report.html` to open automatically.

PowerShell fallback

```
Set-Location '<GUI repository>'
conda activate ai4mbe-gui
python -m tools.rheed_postprocessing_labeling build `
  --session '<local-data-root>\growth_session.zip' `
  --predictions '<local-data-root>\predictions.csv' `
  --model-spec '<local-data-root>\model_spec.json' `
  --output-dir '<local-data-root>\point-event-review'
```

Copy the entire report directory

The report consists of `interactive_report.html`, `images/`, `vendor/`, `run_manifest.json`, and its adjacent `annotations/` sidecar. Copy the entire directory during handoff. Preserve the original prediction CSV and model-spec JSON separately outside the report directory. Review WebP files are display copies only; the original saved-frame bytes remain in the read-only ZIP.

The desktop application always refuses a non-empty output directory. CLI `--overwrite` may replace only a recognized, unmodified tool output. It validates the new inputs and stages a complete replacement first. If the old report changes during the build, replacement stops and the original content is restored. Filesystem roots, repository paths, the current directory, and directories containing inputs are always rejected.

7 Review and complete point events

In brief. Select an Unfinished point, confirm or reject candidates, edit concise semantic labels, choose the representative interval Anchor, identify the reviewer, and explicitly click Complete. Comments are optional.

The `rheed-point-events-v3` schema has three editable change-event sources: `manual`, `auto_capture`, and `posthoc`, plus the separate initial-state audit item. Multiple events may share one timestamp or frame while keeping different IDs. Current events use UUIDs; legacy IDs deterministically include session, source file, source index, time, and capture sequence. The editor shows three tracks: event points, automatically derived full-state intervals, and interval Anchors. Mark In/Out controls are not part of this workflow.

Legacy `rheed-point-events-v2` evidence is validated and displayed read-only. Its deprecated `surface_quality` field is preserved exactly and is never reinterpreted as v3 pattern clarity. Start a separate v3 annotation sidecar if further editing is required.

Immutable original point and editable review point

The original point retains event ID, source, time, source frame when available, capture sequence, any pre-existing image SHA-256 evidence, original note, and source-row SHA-256. It never moves. The review point is the editable event location and can move only by snapping to an actual saved frame. Moving it never changes the original evidence, so the timeline preserves both what was recorded live and where the reviewer finally placed the event.

Concise semantic labels

Each event can contain one or more editable labels:

- **Reconstruction appeared** or **Reconstruction disappeared**: `1x1`, Twinned `2x1`, `c(6x2)`, RT13, or HTR.
- **Pattern clarity became**: Good or Bad.

Use the smallest set that describes the observed change. A point may contain more than one label when different changes occur together. Same-frame changes are applied atomically. Within one event, each reconstruction may occur at most once and with only one direction: appeared or disappeared, never both. Across the run, replay rejects repeated appearance, disappearance of an absent type, and Good-to-Good or Bad-to-Bad changes. Good/Bad describes the visible pattern clarity only. It must not replace image-usability, instrument-validity, chemical surface-quality, or FeSe film-quality records.

Candidate decisions and the initial state

Every automatic-change candidate must be explicitly **Confirmed** or **Rejected**. Confirming accepts the event and requires its semantic label or labels. Rejecting clears all semantic labels and its interval Anchor while retaining the immutable detector source evidence. The rejection revision preserves the complete `before` state for audit and replay. Review the initial-state item separately; it establishes the default first interval and must not be counted as a physical appearance event.

Accepted point changes are replayed from the initial state. Every unique boundary produces a half-open interval whose exported record contains the complete reconstruction-presence set and the current Good/Bad/unknown clarity, rather than only the latest event delta.

Representative interval Anchor

The star-shaped **Anchor** belongs to one derived state interval and selects the saved frame that most clearly represents that complete state. It is not a second event and is exported under the interval. Set or drag it only to an actual saved frame inside that interval. Events at the same time share one following interval and one Anchor. Moving a boundary invalidates an Anchor that no longer lies in its interval and returns the affected review to Draft.

Each interval has exactly one Anchor slot. It may be empty only while the interval is Unfinished; duplicate Anchors are invalid, and Complete requires that the slot identify one actual saved frame inside the interval.

Routine live, posthoc, report, event-movement, and Anchor-selection operations identify a frame by stable session identity, capture sequence, and a saved-frame locator such as archive member, frame index, or frame path. A hash already present may remain as optional fail-closed evidence, but the labeling workflow never computes one solely to create, move, report, or select an annotation. Registered archive and evaluation protocols retain their separate hash-integrity requirements.

Step 1. Select an event in **Unfinished**, or choose **Add posthoc event at playhead** on an actual saved frame.

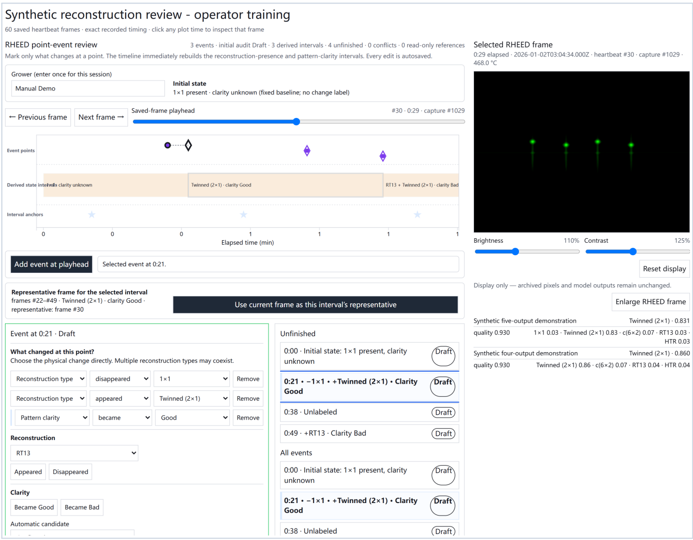
Step 2. Compare the immutable original marker with the movable review marker. Move only when another saved frame is a better review anchor.

Step 3. Review the initial-state item, or choose **Confirmed** or **Rejected** for an automatic candidate.

Step 4. Add, edit, or remove reconstruction and pattern-clarity labels. A confirmed event must retain at least one valid label.

Step 5. Set the representative interval Anchor on an actual saved frame. Enter the reviewer; confidence and comment are optional.

Step 6. Save the Draft and click **Complete** explicitly.



Actual application screenshot. The actual editor using generated demo inputs. Event-point, derived-state, and interval-Anchor tracks plus the Unfinished queue replace hand-drawn interval controls.

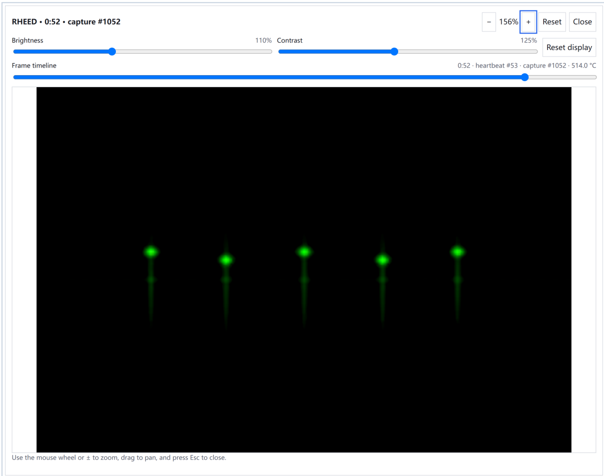
For a confirmed change event, Complete requires a reviewer, at least one valid semantic label, and a representative interval Anchor. The initial-state audit item requires a reviewer and first-interval Anchor but no change label. A rejected automatic candidate requires the reviewer and explicit Rejected decision, but no semantic label or interval Anchor. Comments are optional. Editing completed review content returns it to Draft. Use **Reopen** for intentional further review.

A source event from `manual` or `auto_capture` cannot be deleted. It may be dismissed only with a reason, preserving the evidence. A `posthoc` event can be deleted and restored through audited revisions.

8 Interpret Anchors, logs, and display controls

In brief. The event point and representative interval Anchor have different meanings. Sensor values are read-only context, and brightness, contrast, zoom, and enlargement never modify archived pixels or model outputs.

The draggable event marker sets the reviewed change point. The state track is recalculated from all accepted changes, beginning from the default 1x1 state. The star-shaped Anchor selects the clearest representative frame inside one resulting interval. Moving either control snaps to an actual saved frame and preserves immutable source evidence. The Anchor is training metadata for the interval; it does not create an additional appearance, disappearance, or clarity change. Equalizer, if used elsewhere as a diagnostic, is separate and is not shown or stored as an event-label input.



Actual application screenshot. The actual enlarged-frame dialog. Its timeline changes the selected saved frame while preserving zoom, pan, brightness, and contrast.

The image, playhead, model guides, event markers, representative Anchors, and enlarged timeline remain synchronized. Temperature, voltage, current, pressure, and data age are looked up from the original logs at event time and remain read-only. A common displayed time does not prove physically simultaneous sampling.

9 Preserve revisions, drafts, and exports

In brief. Every edit is append-only. The source ZIP and acquisition CSVs remain unchanged, offline changes live in `annotations/`, and canonical JSON retains more provenance than CSV.

Every revision records actor, UTC, action, complete before/after state, and base revision ID. Live sessions write `rheed_event_revisions.jsonl` plus a rebuildable current-state summary. Offline review writes an adjacent sidecar; it never changes the ZIP. A transaction marker permits exactly-once recovery after interruption.

Browser local storage is only a convenience Draft. Export JSON frequently. CSV is useful for tables and collaboration but does not replace nested semantic labels, candidate decisions, representative Anchors, and revision provenance in JSON.

```
python -m tools.rheed_postprocessing_labeling validate `
  --report '<local-data-root>\point-event-review\interactive_report.html' `
  --annotations '<local-data-root>\point-event-review\annotations\point_events.json'
```

Validation binds the archive, ordered frame hashes, model context, event IDs, immutable source evidence, review points, representative interval Anchors, semantic labels, candidate decisions, revision chain, and Complete gate. Never bypass a mismatch. Old `rheed-temporal-segments-v1` files remain available to their original validator as read-only compatibility data; do not convert segment start, middle, or end into a point automatically.

10 Data safety and scientific boundaries

In brief. Keep real laboratory data and outputs out of Git. This workflow is model-assisted review of bare STO reconstruction, not blind-gold truth, acquisition-quality labeling, or FeSe film-quality evaluation.

May be tracked	Keep out by default
Tool code, templates, tests, and documentation	Real ZIP files, raw RHEED images, and sensor logs
Example model specifications	Predictions, generated reports, annotations, and checkpoints
Sanitized screenshots made with generated inputs	Credentials, unpublished results, and experiment screenshots

Model plots are visible during this workflow, so its reviews are model-assisted and not blind gold. Model argmax must not overwrite a human label. Image acquisition quality means whether a captured frame can be analyzed and is separate from **Pattern clarity became Bad**. Neither is evidence of poor chemical surface quality or film quality. The current reconstruction target is bare STO before growth, not FeSe film quality.

Preserve the read-only source ZIP and SHA-256, report manifest, revision journal, canonical JSON, model specifications, and complete report directory during handoff.

11 Troubleshoot without destroying evidence

In brief. Record the exact error, time, chamber, launcher, commit, environment, paths, event ID, and validity state before changing anything. Never copy settings between chambers or hand-edit evidence.

Symptom	Safe action
Shortcut missing, old checkout, or old icon	Run the shortcut installer from the intended current checkout; verify the shared launcher path and bundled icon
Wrong chamber title	Close normally and use the matching chamber launcher
Optional live-driver warning	Preserve the launcher log and do not ARM a production mode that needs the missing driver
Direct exposure refused	Stay idle; for a nonzero request verify Full camera access and release kSA or Vimba X Viewer; never bypass readback or restoration checks
START disabled or frame counter stalled	DISARM; inspect camera ownership, triggering, capture sequence, arrival age, and the exact worker error
Instrument value invalid	Record mode, connected/valid/error state, sequence, and age; do not guess COM or set-point values
Build rejects inputs	Check ZIP, prediction/spec pairs, hashes, and an empty output directory
HTML has no images	Restore the complete report directory
Unfinished event missing	Verify source CSV row and row hash; preserve revision replay errors
Automatic candidate remains Pending	Inspect it and explicitly choose Confirmed or Rejected; never infer the decision from detector score
Representative Anchor refused	Move it to a saved frame inside the stable interval before the next active, non-rejected semantic event
Complete disabled	Supply the reviewer and, for a confirmed event, a semantic label plus representative Anchor; a rejected candidate needs an explicit Rejected decision
Revision recovery fails	Preserve JSONL and the pending marker; do not edit either file

For O-MBE, confirm **Oxide MBE Growth Monitor** and use only the O-MBE startup record. For Ch-MBE, confirm **Chalcogenide MBE Growth Monitor** and use only the Ch-MBE startup record. Preserve `git status --short --branch`, `git rev-parse HEAD`, launcher logs, and full validation messages.

12 Verify versions and finish the checklist

In brief. Record versions and hashes, complete only the selected chamber checklist, resolve or dismiss every Unfinished event, and validate the canonical point-event export.

```
Set-Location '<GUI repository>'
git status --short --branch
git rev-parse HEAD
conda activate ai4mbe-gui
python --version
python -m tools.rheed_postprocessing_labeling --help
Get-FileHash '<local-data-root>\growth_session.zip' -Algorithm SHA256
```

O-MBE startup checklist

- ☐ Use `Start O-MBE Growth Monitor.cmd` and confirm **Oxide MBE Growth Monitor**.
- ☐ Confirm launcher repository, branch, commit, Python, passed chamber preflight, and optional-driver status.
- ☐ Confirm `vimba / modbus / ads / elog` against the O-MBE record.
- ☐ Confirm the O-MBE direct-exposure request, Full-access requirement when writing, and reported readback; never borrow Ch-MBE's exposure.
- ☐ Confirm advancing RHEED capture sequence and a current frame, valid instrument states, data age, and log directory.
- ☐ ARM only under the O-MBE SOP and authorized operator.

Ch-MBE startup checklist

- ☐ Use `Start Ch-MBE Growth Monitor.cmd` and confirm **Chalcogenide MBE Growth Monitor**.
- ☐ Confirm launcher repository, branch, commit, Python, passed chamber preflight, and optional-driver status.
- ☐ Confirm `vimba / modbus / ads / elog` against the Ch-MBE record.
- ☐ Confirm the Ch-MBE direct-exposure request, Full-access requirement when writing, and reported readback; never borrow O-MBE's exposure.
- ☐ Confirm advancing RHEED capture sequence and a current frame, valid instrument states, data age, and log directory.
- ☐ ARM only under the Ch-MBE SOP and authorized operator.

Shared offline checklist

- ☐ Confirm session metadata names the intended chamber and preserve the ZIP hash.
- ☐ Open through the desktop labeler for durable revisions and Complete.
- ☐ Resolve every Unfinished item or dismiss it with a documented reason.
- ☐ Confirm every automatic candidate has an explicit Confirmed or Rejected decision; the fixed initial state is not a candidate.
- ☐ Complete the initial-state audit with its reviewer and first-interval Anchor; do not add an appearance label.
- ☐ Confirm each Complete accepted event has reviewer, semantic label, representative interval Anchor, and revision history; comments remain optional.
- ☐ Export JSON, validate it, and hand off the complete report directory.

Searchable manual: `docs/RHEED_GUI_Postprocessing_Labeling_User_Manual_EN.md`

AI prompt pack: `docs/RHEED_GUI_Postprocessing_Labeling_AI_Prompt_Pack_EN.md`

Stop and contact the maintainer when instrument state, stale data, version identity, provenance, event recovery, or validation cannot be explained.

Application screenshots were captured from the actual software using generated demo inputs; they contain no experimental data.